JobPilot AI - Job Domains, Titles and Assessment Matrix

Reference document for the recruitment-automation pipeline of an IT company. It lists job domains, the job titles inside each domain, and which assessment each role should receive at every stage. Intended to be read by Claude Code and used as the single source of truth for assessment assignment. A machine-readable copy is in **role_assessment_matrix.json**.

1. Pipeline model

Every candidate goes through **4 stages**. Stages 1 and 4 are identical for every role. Stages 2 and 3 are **chosen per job title**, because aptitude maths and programming do not fit every role.

Stage	Applies to	Technical roles	Non-technical roles
S1 Self-intro	All	Same	Same
S2 Screening test	Per role	Aptitude (Core / Logic)	Situational Judgement (SJT), or Numerical aptitude for finance/data-like roles
S3 Practical test	Per role	Programming, light scripting/SQL, hands-on lab, or test-design	Work sample: written, data/spreadsheet, design, role-play, or case study
S4 AI interview	All	Same engine, deep technical questions	Same engine, scenario/behavioural questions

Key rule: programming is replaced by a *job-realistic work sample* for non-coding roles. This is the same idea used by pre-employment platforms: cognitive aptitude tests, situational judgement tests and work-sample tests are mixed by job family (TestGorilla, Testlify, Xobin, WeCreateProblems; see sources).

2. Assessment catalogue

Code	Name	What it is
S1-INTRO	Self-Introduction (all roles)	Candidate records a 60-120 sec video/audio intro. Scored on clarity, communication, confidence, relevance to role. Same for every role.
AP-CORE	Aptitude - Core	Quantitative + logical reasoning + verbal, 25-30 Q, 30 min. For roles that need analytical/engineering thinking.
AP-LOGIC	Aptitude - Logic/Verbal	Logical + verbal + pattern reasoning, 20 Q, 20 min, light maths. For support/ops-leaning tech roles.
AP-NUM	Aptitude - Numerical/Data	Number series, percentages, ratios, chart/table interpretation, 20-25 Q, 25 min. For finance/data-heavy roles.
SJT	Situational Judgement Test	10-15 workplace scenarios, pick best/worst action, 20 min. For people-facing/non-technical roles where aptitude math adds little.
PRG	Programming Test	2-3 coding problems (easy/medium) in an online IDE with test cases, 60-90 min. Language chosen by role.
PRG-L	Light Programming / SQL / Scripting	1-2 tasks: SQL queries, Python/Bash script, or config-as-code, 45 min.
TECH-LAB	Hands-on Technical Scenario	Domain MCQ (15) + troubleshooting/lab scenario (log reading, config fix, architecture or incident question), 45-60 min. No heavy coding.
TEST-DESIGN	Test Design Task	Given a feature/spec: write test cases, find seeded bugs, file a bug report, 45 min.
WS-WRITE	Written Work Sample	Job-realistic writing task (email, brief, post, policy note, report) graded by rubric + AI, 30-45 min.
WS-DATA	Data/Spreadsheet Work Sample	Provided messy dataset: clean, calculate, summarise findings (Excel/Sheets-style), 45 min.
WS-DESIGN	Design Task	Short design brief: wireframe/critique/flow or portfolio case review, 60 min or take-home 24-48 h.
WS-ROLE	Role-play Simulation	Simulated chat/email/call with a customer or prospect (AI persona), scored on empathy, accuracy, process, 15-20 min.
CASE	Case Study / Prioritisation	Business case: prioritise, plan, estimate, recommend, 45 min written + short defence in interview.
S4-AI-INT	Final AI Interview (all roles)	Live AI voice interview, 15-30 min. Question bank is generated per job title + level (technical depth for tech roles; scenario/behavioural for non-tech).

A dash (-) in a stage means the stage is waived (senior/leadership hires, where experience and interview carry the weight).

3. Assignment rules (for the code)

• Look up the job title in the matrix. If found, use its S2 and S3 codes. If the title is new, match it to the closest domain and use the domain default.

• Technical roles that write code daily (Software, QA automation, Data engineering, ML) always get **PRG** at S3 and **AP-CORE** at S2.

• Technical roles that mostly operate/diagnose systems (DevOps, IT support, security, enterprise apps) get **TECH-LAB**; code only if scripting is in the JD.

• Hybrid roles (Product, Design, Business Analyst) need tech literacy but no coding: use **CASE**, **WS-DESIGN** or **WS-WRITE**.

• Non-technical people-facing roles get **SJT** at S2; number-heavy ones (finance, payroll, SEO, procurement) get **AP-NUM**.

• Seniority modifier: Intern/Fresher - raise aptitude weight, easier S3. Senior (5+ yrs) - shorter S2, harder S3. Leadership - waive S2, S3 = CASE.

• Suggested pass marks: S2 60%, S3 60-70% (70% for coding), S4 scored by rubric (communication, role knowledge, problem solving, integrity). Tune after the first real cohort.

• Integrity: randomised question order, timed sections, tab-switch/paste logging, per-candidate generated variants for work samples, AI-assisted plagiarism check on written tasks.

4. Domains, job titles and assigned assessments

Every row below also gets S1-INTRO first and S4-AI-INT last. Columns show S2 and S3. Key focus is what the S3 test and the AI interview should emphasise.

4.1 Software Development (Technical)

Core engineering roles. Programming language picked from the job description.

Job title	S2	S3	Key focus
Frontend Developer	AP-CORE	PRG	JavaScript/TypeScript, React/Angular/Vue, DOM, CSS, UI logic
Backend Developer	AP-CORE	PRG	Java/Python/Node/.NET/Go, APIs, DB queries, data structures
Full-Stack Developer	AP-CORE	PRG	Frontend + backend task in one problem set
Mobile App Developer (Android/iOS/Flutter)	AP-CORE	PRG	Kotlin/Swift/Dart, lifecycle, state, API integration
Embedded / Firmware Engineer	AP-CORE	PRG	C/C++, memory, bit ops, protocols
Game Developer	AP-CORE	PRG	C++/C#, algorithms, maths/physics logic
Software Engineer Trainee / Intern	AP-CORE	PRG	Easier problem set; DSA basics, weight aptitude higher
Software Architect / Principal Engineer	-	CASE	System design case instead of coding; S2 waived

4.2 Quality Assurance & Testing (Technical)

Manual testers get test-design; automation roles get coding.

Job title	S2	S3	Key focus
Manual QA Tester	AP-LOGIC	TEST-DESIGN	Test cases, bug reports, edge cases, SDLC/STLC
QA Automation Engineer	AP-CORE	PRG	Selenium/Playwright/Cypress + scripting in Java/Python/JS
SDET	AP-CORE	PRG	Coding + framework design + API testing
Performance Test Engineer	AP-LOGIC	PRG-L	JMeter/k6 scenario design, metrics reading, scripting
QA Lead / Test Manager	-	CASE	Test strategy case; S2 optional

4.3 DevOps, Cloud & Infrastructure (Technical)

Hands-on troubleshooting beats algorithms for these roles; scripting task optional.

Job title	S2	S3	Key focus
DevOps Engineer	AP-LOGIC	TECH-LAB	CI/CD, Docker, Kubernetes, Linux, IaC, Git
Site Reliability Engineer (SRE)	AP-CORE	PRG-L	Incidents, monitoring, SLOs, Python/Go scripting
Cloud Engineer (AWS/Azure/GCP)	AP-LOGIC	TECH-LAB	Networking, IAM, compute/storage, cost, architecture scenario
Platform Engineer	AP-LOGIC	TECH-LAB	Internal tooling, Kubernetes, Terraform
System Administrator	AP-LOGIC	TECH-LAB	Linux/Windows server, users, backups, patching
Network Engineer	AP-LOGIC	TECH-LAB	TCP/IP, routing, switching, firewall, subnetting
Database Administrator (DBA)	AP-LOGIC	PRG-L	SQL tuning, backup/recovery, indexing, replication

4.4 Data, AI & Machine Learning (Technical)

SQL/Python-centred. Analyst roles lean on a data work sample; ML roles on modelling questions.

Job title	S2	S3	Key focus
Data Analyst	AP-NUM	WS-DATA	SQL, Excel, dashboards, insight write-up
Business Intelligence (BI) Developer	AP-NUM	PRG-L	SQL, Power BI/Tableau, data modelling
Data Engineer	AP-CORE	PRG	SQL + Python/Scala, ETL, pipelines, warehouses
Analytics Engineer	AP-CORE	PRG-L	SQL, dbt, modelling, testing
Data Scientist	AP-CORE	PRG-L	Python, statistics, feature work, evaluation metrics
Machine Learning Engineer	AP-CORE	PRG	Python coding + ML concepts, deployment/MLOps
AI / LLM Engineer	AP-CORE	PRG	Prompting, RAG, APIs, evaluation, Python
Data Annotation / AI Trainer	AP-LOGIC	WS-WRITE	Labelling guidelines, accuracy, language quality

4.5 Cybersecurity (Technical)

Scenario-based labs: log analysis, vulnerability triage, threat modelling.

Job title	S2	S3	Key focus
SOC Analyst	AP-CORE	TECH-LAB	SIEM logs, alert triage, incident steps
Penetration Tester / Ethical Hacker	AP-CORE	TECH-LAB	OWASP Top 10, recon, exploitation reasoning, reporting
Security Engineer	AP-CORE	PRG-L	Secure coding review, cloud security, scripting
GRC / Compliance Analyst (ISO 27001, SOC 2)	SJT	CASE	Policy, risk register, audit scenario - no hands-on lab
Identity & Access Management Specialist	AP-CORE	TECH-LAB	SSO, MFA, RBAC, directory services

4.6 IT Support & Service Desk (Technical)

Troubleshooting scenarios plus a short communication check.

Job title	S2	S3	Key focus
Help Desk / Service Desk Analyst (L1)	SJT	TECH-LAB	Common faults, ticket handling, communication
Technical Support Engineer (L2/L3)	AP-LOGIC	TECH-LAB	Log reading, root cause, product/API knowledge
Desktop / End-User Support Technician	SJT	TECH-LAB	OS, hardware, printers, VPN, Active Directory
IT Asset / Systems Coordinator	SJT	WS-DATA	Inventory spreadsheets, procurement tracking

4.7 Enterprise Applications & Automation (Technical)

Platform-specific consultants and low-code/automation developers.

Job title	S2	S3	Key focus
SAP / ERP Consultant	AP-LOGIC	TECH-LAB	Module knowledge (FI/MM/SD), configuration scenarios
Salesforce Developer / Administrator	AP-LOGIC	TECH-LAB	Objects, flows, Apex basics, security model
ServiceNow Developer	AP-LOGIC	PRG-L	Workflows, scripting (JavaScript), ITSM
RPA Developer (UiPath/Power Automate)	AP-LOGIC	TECH-LAB	Process automation design, selectors, exceptions
Low-Code / No-Code Developer	AP-LOGIC	TECH-LAB	App building scenario, data model, integrations

4.8 Product & Project Management (Hybrid)

Hybrid: tech-literate but not coding. Case study replaces programming.

Job title	S2	S3	Key focus
Product Manager	AP-CORE	CASE	Prioritisation, metrics, roadmap, trade-offs
Product Owner	AP-CORE	CASE	Backlog, user stories, acceptance criteria
Project / Delivery Manager	SJT	CASE	Plan, risk, stakeholder, estimate
Scrum Master / Agile Coach	SJT	CASE	Ceremonies, impediments, team conflict
Business Analyst	AP-CORE	WS-WRITE	Requirements doc / user story writing from a messy brief
Technical Program Manager	AP-CORE	CASE	Cross-team dependency and system-level case

4.9 UI/UX & Design (Hybrid)

Portfolio + design task replaces aptitude-heavy screening.

Job title	S2	S3	Key focus
UI Designer	SJT	WS-DESIGN	Layout, typography, colour, component consistency
UX Designer / Product Designer	SJT	WS-DESIGN	User flow, wireframe, usability critique
UX Researcher	SJT	CASE	Research plan, interview guide, synthesis
Graphic / Visual Designer	SJT	WS-DESIGN	Brand asset task, composition, tool proficiency
Motion / Video Designer	SJT	WS-DESIGN	Short storyboard/animation brief
Technical Writer	AP-LOGIC	WS-WRITE	Documentation, API doc clarity, editing

4.10 Sales & Business Development (Non-technical)

Role-play (cold outreach, discovery call, objection handling) is the strongest predictor.

Job title	S2	S3	Key focus
Sales Development Rep (SDR)	SJT	WS-ROLE	Cold email/call, qualification, persistence
Account Executive	SJT	WS-ROLE	Discovery, demo pitch, negotiation role-play
Business Development Manager	SJT	CASE	Market/account plan, pipeline strategy
Account Manager	SJT	WS-ROLE	Renewal, upsell, escalation handling
Pre-Sales / Solutions Consultant	AP-LOGIC	TECH-LAB	Technical demo, requirement mapping, RFP answer
Inside Sales / Telesales Executive	SJT	WS-ROLE	Call script, objection handling

4.11 Marketing & Content (Non-technical)

Written/creative sample; analytical marketers get a data sample.

Job title	S2	S3	Key focus
Digital Marketing Executive	SJT	WS-WRITE	Campaign copy, channel mix, basic metrics
SEO / SEM Specialist	AP-NUM	WS-DATA	Keyword/data sheet analysis, audit findings
Content Writer / Copywriter	SJT	WS-WRITE	Blog/landing page/email under brief + tone
Social Media Manager	SJT	WS-WRITE	Content calendar, post copy, crisis response
Growth / Performance Marketer	AP-NUM	WS-DATA	Funnel metrics, CAC/ROAS calc, budget split
Brand / PR Executive	SJT	WS-WRITE	Press note, brand voice, stakeholder messaging

4.12 Customer Success & Support (Non-technical)

Simulated chat/email/call with an AI customer; score empathy, accuracy, policy.

Job title	S2	S3	Key focus
Customer Support Executive (Chat/Email)	SJT	WS-ROLE	Reply quality, tone, grammar, speed
Voice / Call Centre Executive	SJT	WS-ROLE	Spoken-English clarity, de-escalation
Customer Success Manager	SJT	CASE	Account health, churn risk plan, QBR outline
Onboarding / Implementation Specialist	SJT	WS-WRITE	Customer onboarding plan and instructions

4.13 Human Resources & Talent (Non-technical)

Policy/people judgement. Recruiters also do a sourcing/screening exercise.

Job title	S2	S3	Key focus
Technical Recruiter / Talent Acquisition	SJT	CASE	Screen resumes vs a JD, write outreach, sourcing plan
HR Generalist / Executive	SJT	WS-WRITE	Policy note, employee query reply, process
HR Business Partner	SJT	CASE	Org/people scenario, advisory recommendation
Learning & Development Specialist	SJT	WS-WRITE	Training outline, needs analysis
Payroll / HR Operations Executive	AP-NUM	WS-DATA	Payroll sheet accuracy, compliance checks
Talent Sourcer / Research Associate	SJT	WS-DATA	Boolean search, candidate sheet building

4.14 Finance & Accounting (Non-technical)

Numerical reasoning + Excel work sample; accuracy is the key signal.

Job title	S2	S3	Key focus
Accountant / Accounts Executive	AP-NUM	WS-DATA	Journal entries, reconciliation, GST/VAT basics
Financial Analyst / FP&A;	AP-NUM	CASE	Budget vs actual, variance commentary
Accounts Payable / Receivable Executive	AP-NUM	WS-DATA	Invoice matching, ageing report
Billing / Revenue Operations Analyst	AP-NUM	WS-DATA	Billing sheet, MRR calc, discrepancy finding
Internal Auditor	AP-NUM	CASE	Control gap case, sampling logic

4.15 Operations, Admin & Legal (Non-technical)

Organisation, communication and accuracy.

Job title	S2	S3	Key focus
Office / Admin Executive	SJT	WS-WRITE	Scheduling, email etiquette, document handling
Executive Assistant	SJT	WS-WRITE	Calendar conflict, prioritisation, drafted reply
Procurement / Vendor Management Executive	AP-NUM	WS-DATA	Quote comparison, vendor scoring sheet
Operations Coordinator	SJT	CASE	Process improvement case
Legal / Contract Executive	AP-LOGIC	WS-WRITE	Clause review, summary of a contract extract
Facilities / Receptionist	SJT	WS-ROLE	Visitor/caller handling simulation

4.16 Engineering Leadership (Technical (senior))

Experienced hires: S2 waived; S3 = system-design / people-management case. Interview carries the weight.

Job title	S2	S3	Key focus
Tech Lead	-	CASE	Design review + code-review exercise
Engineering Manager	-	CASE	Team scenarios, delivery trade-offs
Head / Director of Engineering	-	CASE	Strategy, hiring, budget case
CTO / VP Engineering	-	CASE	Executive panel; AI interview optional

5. Suggested JSON shape for the app

```
{ "domain": "Finance & Accounting", "title": "Financial Analyst / FP&A",
  "stage1": "S1-INTRO", "stage2": "AP-NUM", "stage3": "CASE", "stage4": "S4-AI-INT",
  "focus": "Budget vs actual, variance commentary" }
```

6. Notes and sources

The matrix is a design recommendation built from the public hiring-test practice below and common IT-company job families; company-specific cut-offs and content were not found publicly and must be tuned with your own hiring data. Role lists are representative, not exhaustive.

• Testlify - common types of pre-employment tests: testlify.com/common-types-of-pre-employment-tests-and-their-uses

• Xobin - types of pre-employment assessments: xobin.com

• WeCreateProblems - 12 types of pre-employment assessment tests: wecreateproblems.com/blog/types-of-pre-employment-assessment-tests

• easy.jobs - types of assessment tests for jobs: easy.jobs/blog/types-of-assessment-tests-for-jobs

• G2 - 9 pre-employment tests: g2.com/articles/pre-employment-testing

• TestGorilla - role-specific skills tests (QA, DevOps, Data Scientist, Product Manager, Support): testgorilla.com

• KPMG / Aon hi-tech job family matrices (job titles by family): assets.kpmg.com, aon.com